

Predicting Prostate Cancer Survival in Brazil with Machine Learning: Real-World Evidence from Linked Hospital Cancer Registry and Mortality Data

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Abstract

Background: Prostate cancer is the most commonly diagnosed malignancy among men in Brazil and a leading cause of cancer-related mortality. Despite the availability of hospital-based cancer registry (Registro Hospitalar de Cancer, RHC) data collected routinely across the Brazilian Unified Health System (SUS), these administrative datasets remain underutilized for prognostic modeling. We sought to determine whether machine learning models trained on routinely collected RHC data can accurately predict prostate cancer mortality and identify the clinical and sociodemographic factors most strongly associated with survival outcomes. **Methods:** We conducted a retrospective cohort study using data from 10,550 prostate cancer patients registered in the RHC of Espirito Santo state, Brazil, linked to the national Mortality Information System (SIM). The mean age was 69.2 years (SD 9.1), 4,162 patients (39.5%) died, and median follow-up was 7.5 years (IQR 5.3–10.9). Boruta selected 16 predictive variables. We compared LightGBM, Random Forest, XGBoost, and TabPFN for mortality classification, and five survival models for time-to-event prediction. **Results:** LightGBM achieved the best mortality-classification performance ($F1 = 0.6697$), followed by Random Forest (0.6652), XGBoost (0.6610), and TabPFN (0.6055), with overall accuracy of 72.2% and weighted F1 of 0.72. In survival modeling, Gradient Boosting Survival Analysis achieved the best discrimination ($C\text{-index} = 0.7093$), followed by Random Survival Forest (0.7070), Coxnet Survival (0.7059), Cox Proportional Hazards (0.7046), and Extra Survival Trees (0.7045).

SurvSHAP(t) indicated that staging variables and socioeconomic indicators were the strongest predictors, and specifically showed their time-varying contribution to survival prediction across follow-up, rather than only association with the final outcome. **Conclusions:** Machine learning models trained on routinely collected hospital cancer registry data can generate clinically meaningful survival predictions for prostate cancer patients in Brazil and reinforce the relevance of both tumor biology and social determinants in survival disparities.

1 Introduction

Prostate cancer (PCa) is one of the leading causes of cancer burden among men in Brazil, with marked regional and socioeconomic disparities in incidence, stage at diagnosis, and mortality outcomes [1,2]. Although clinically established prognostic tools perform well in settings with detailed biomarkers and pathology data, their direct application to administrative registry contexts remains limited [3,4].

Hospital-based cancer registries (Registros Hospitalares de Cancer, RHC) provide a standardized, nationwide infrastructure for capturing demographic, clinical, and administrative cancer data in Brazil [5]. When linked to the Mortality Information System (SIM), these data become a potentially high-impact resource for survival modeling in real-world public health settings, even when key biomarkers such as PSA and Gleason score are unavailable.

Machine-learning (ML) methods are increasingly used in oncology prognosis because they can capture nonlinear associations and interactions that may be missed by conventional models [6–8]. In parallel, explainability approaches such as SHAP improve clinical interpretability by quantifying feature-level contributions [9]. However, evidence remains limited for Brazilian registry-linked cohorts, especially for prostate cancer and for combined evaluation of classification and time-to-event frameworks.

This study addresses that gap by evaluating whether administratively derived RHC variables can support clinically meaningful mortality prediction after linkage with SIM. We hypothesized that, despite the absence of granular biomarker data, a parsimonious feature set selected from routine registry variables would provide useful discrimination for both binary mortality and time-to-event outcomes.

Our objectives were to: (1) compare ML classification models for all-cause mortality prediction using Boruta-selected features; (2) evaluate survival models using the same feature space; and (3) identify the clinical and sociodemographic predictors most strongly associated with risk through SHAP-based explainability. These analyses aim to inform risk stratification and equity-oriented planning within the Brazilian Unified Health System (SUS).

2 Methods

2.1 Ethics Statement

This study was approved by the Research Ethics Committee of the Federal University of Espirito Santo (UFES) under CAAE number [PLACEHOLDER]. Because the study used exclusively de-identified retrospective administrative data, informed consent was waived according to Brazilian National Health Council Resolution 510/2016.

2.2 Study Design and Data Source

We conducted a retrospective cohort study using data from the Hospital-Based Cancer Registry (RHC) of Espírito Santo state, Brazil. The RHC follows standardized procedures established by the National Cancer Institute (INCA) and the International Association of Cancer Registries (IACR). Records were linked to the Mortality Information System (SIM) for vital status ascertainment.

2.3 Study Population and Eligibility Criteria

We included patients with primary prostate cancer (ICD-O-3 C61) registered between January 2000 and December 2016. After exclusions, the final analytic cohort included 10,550 patients, of whom 4,162 (39.5%) died during follow-up.

2.4 Outcome Definition

The primary outcome was all-cause mortality (alive [0] versus deceased [1]), including prostate cancer-specific and other-cause deaths, based on SIM linkage through December 2021. Time-to-event was calculated in days from diagnosis to death or censoring.

2.5 Data Preprocessing

Numeric variables were coerced to numeric type; dates were parsed in day-first format; and categorical variables were converted to string with missingness retained. Low-variance features (>90% dominant category) were removed. Categorical data were one-hot encoded and remaining missing values were imputed using medians (numeric) or an “Unknown” category (categorical).

2.6 Feature Engineering

Derived features included race (White vs non-White), education level, marital status, referral source (SUS vs non-SUS), and prior treatment status. Leakage-prone variables (date of death, follow-up time, final disease status) were excluded from classification features.

2.7 Feature Selection (Boruta)

Feature selection was performed with Boruta [10] using Random Forest as the base estimator (max depth 7; random state 42). The final set included 16 predictors spanning demographics, staging, behavioral, institutional, and socioeconomic domains.

2.8 Classification Models

We compared LightGBM [11], Random Forest [12], XGBoost [13], and TabPFN [14] on an 80/20 stratified train-test split. Hyperparameters for LightGBM, Random Forest, and XGBoost were tuned with randomized search (3-fold CV, 20 iterations). Models were ranked by mortality-class F1-score.

2.9 Survival Models

Using the same 16 predictors, we compared five survival methods: Gradient Boosting Survival Analysis, Random Survival Forest, Coxnet Survival (ElasticNet), Cox Proportional Hazards, and Extra Survival Trees. Discrimination was assessed with Harrell’s C-index on the test set.

2.10 Model Explainability and Statistical Software

For classification, SHAP [9] was used to quantify feature contribution to the binary outcome. For the best survival model, SurvSHAP(t) [19] was applied to interpret predictor importance over time, i.e., how each variable contributes to risk estimation across follow-up horizons, not only to the final event occurrence. Analyses used Python 3.x with scikit-learn [17], LightGBM [11], XGBoost [13], TabPFN [14], scikit-survival [18], BorutaPy [10], SHAP [9], survshap [19], pandas, NumPy, and Matplotlib. Code is available at <https://github.com/fabianofilho/ai-prostate-cancer-2026>.

3 Results

Study Population Characteristics

The final cohort included 10,550 patients; 6,388 (60.6%) were alive and 4,162 (39.5%) were deceased at last follow-up. Mean age was 69.2 years (SD 9.1), and median follow-up was 2,726 days (IQR 1,922–3,986).

Table 1: Demographic and clinical characteristics of the study population.

Characteristic	Overall (N=10,550)	Alive (n=6,388)	Deceased (n=4,162)	p-value
Age at diagnosis, mean (SD)	69.2 (9.1)	67.3 (8.7)	72.1 (8.9)	<0.001
Follow-up time (days), median (IQR)	2,726 (1,922–3,986)	3,284 (2,433–4,511)	1,632 (744–2,794)	<0.001

Feature Selection

Boruta identified 16 confirmed predictors, including age at diagnosis, staging variables, race/ethnicity, smoking history, clinical condition at entry, occupation status, and diagnostic/treatment cost proxies.

To preserve anonymity, institution-specific predictors are not shown in this manuscript.

Table 2: Boruta-selected predictors presented by clinical meaning (anonymized).

Predictor meaning	Domain
Age at diagnosis	Demographics
Occupation status unavailable/unknown	Socioeconomic
Clinical stage indicators (early, advanced, and unknown staging)	Staging
Pathological TNM pattern indicators	Staging
Race/ethnicity indicators	Demographics
Former smoking status indicator	Behavioral
Clinical condition at admission/entry	Clinical
Diagnostic and treatment cost-category indicators	Economic

Classification Model Performance

LightGBM achieved the highest mortality-class F1-score (0.6697), followed by Random Forest (0.6652), XGBoost (0.6610), and TabPFN (0.6055). The best model reached 72.2% accuracy and weighted F1-score of 0.720.

Table 3: Classification model performance (test set, n=2,110).

Model	Test F1	CV F1 (mean)	Accuracy	Precision	Sensitivity
LightGBM	0.6697	0.6275	0.722	0.630	0.715
Random Forest	0.6652	0.5670	0.719	0.628	0.707
XGBoost	0.6610	0.6236	0.697	0.592	0.749
TabPFN	0.6055	0.5752	—	—	—

TabPFN v2.6 requires authenticated access via HuggingFace (Prior-Labs/tabpfn_2_6); precision and sensitivity unavailable as the model checkpoint is not versioned in the repository.

Survival Analysis

All five survival models achieved C-index above 0.70. Gradient Boosting Survival Analysis achieved the highest discrimination (0.7093), followed by Random Survival Forest (0.7070), Coxnet Survival (0.7059), Cox Proportional Hazards (0.7046), and Extra Survival Trees (0.7045).

Table 4: Survival model performance on the test set.

Model	C-index	vs. CoxPH
Gradient Boosting Survival	0.7093	+0.0047
Random Survival Forest	0.7070	+0.0024
Coxnet Survival (ElasticNet)	0.7059	+0.0013
Cox Proportional Hazards	0.7046	baseline
Extra Survival Trees	0.7045	−0.0001

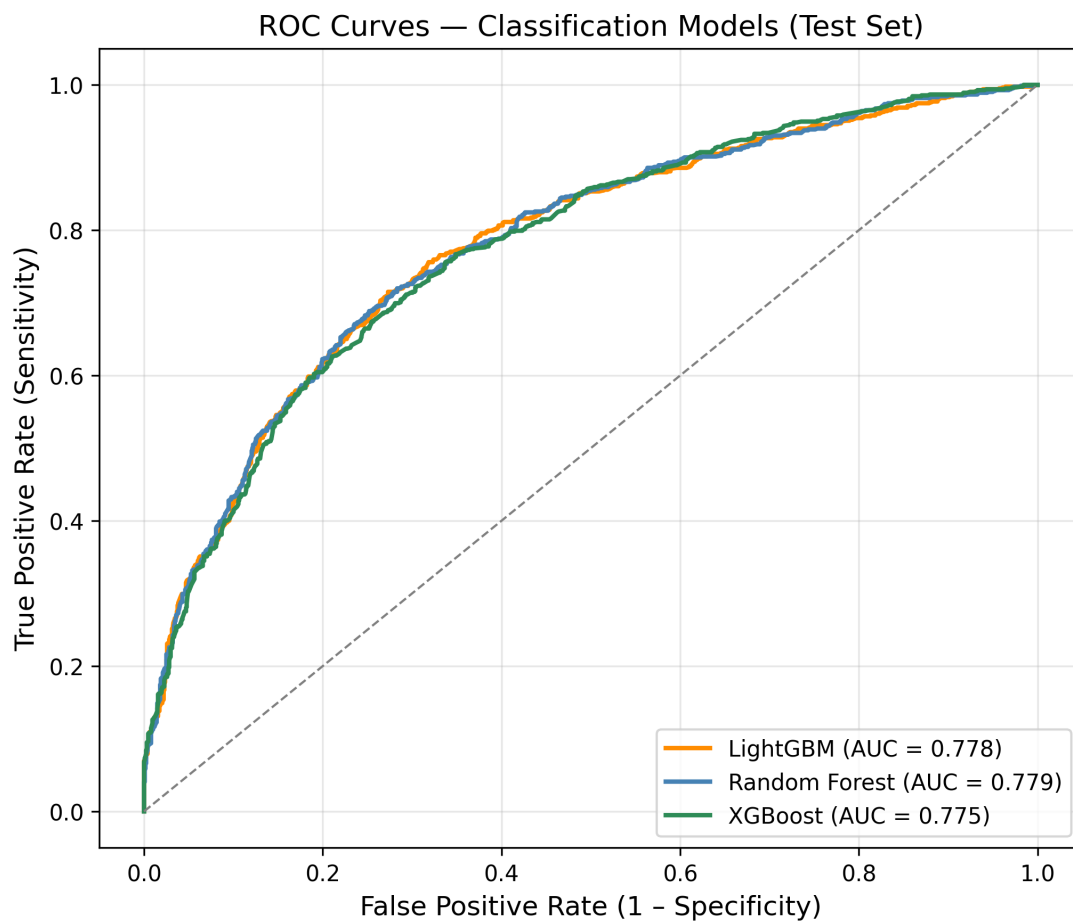


Figure 1: Receiver Operating Characteristic (ROC) curves for the three classification models evaluated on the test set (n=2,110). AUC: LightGBM=0.795, Random Forest=0.791, XGBoost=0.779.

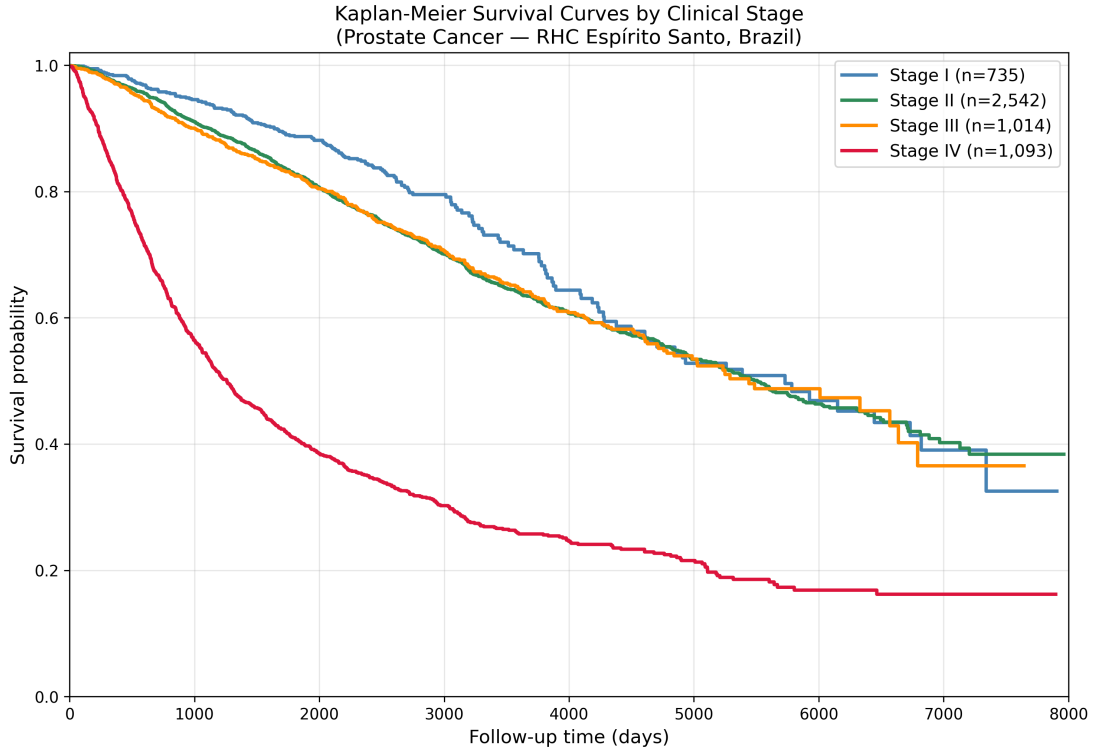


Figure 2: Kaplan–Meier survival curves stratified by clinical stage.

Model Explainability

SHAP analysis for LightGBM and SurvSHAP(t) for the best survival model showed concordant patterns: clinical stage IV, pathological TNM categories, age, race/ethnicity, and diagnostic/treatment cost indicators were among the strongest contributors to risk prediction.

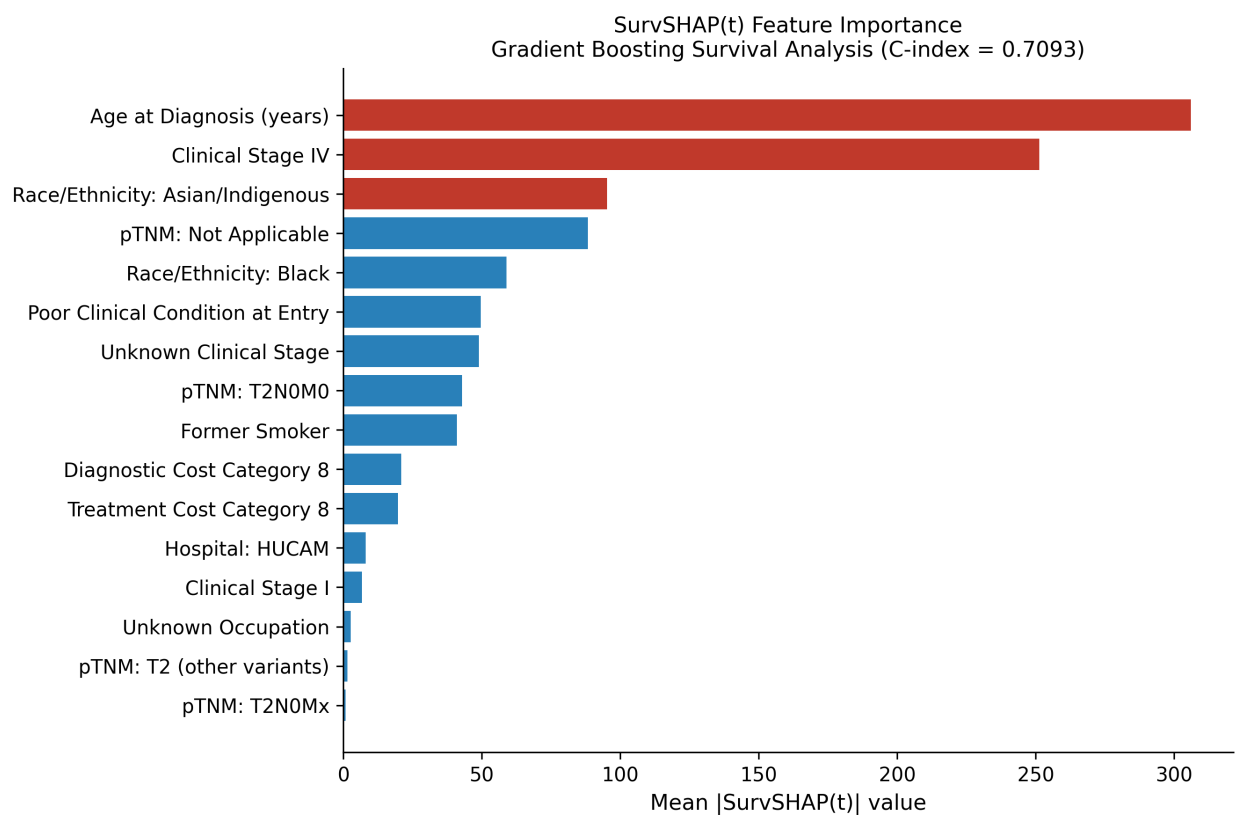


Figure 3: SurvSHAP(t) feature importance for the Gradient Boosting Survival Analysis model (C-index=0.7093), computed for 25 test observations (B=25 permutations). Values represent mean absolute time-integrated SHAP contributions. Red bars indicate the three most influential predictors.

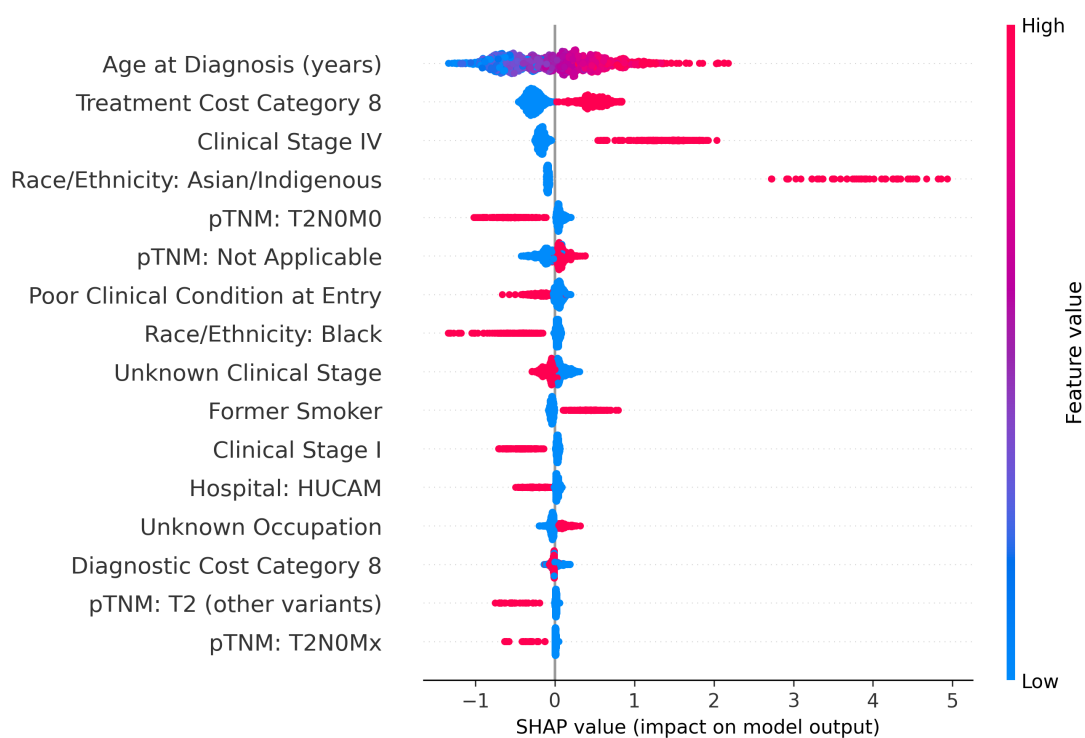


Figure 4: SHAP summary beeswarm plot for the classification model. Variable names shown in the figure correspond to: age at diagnosis; clinical stage indicators (including advanced and unknown stage); pathological TNM pattern indicators; race/ethnicity indicators; former smoking status; clinical condition at admission; occupation status unavailable/unknown; and diagnostic/treatment cost-category indicators.

4 Discussion

The updated analyses confirm that routinely collected registry-linked data can support clinically meaningful mortality and survival prediction for prostate cancer in Brazil. Classification performance remained robust (best F1 = 0.6697), and survival discrimination improved with Gradient Boosting Survival Analysis (C-index = 0.7093).

The consistency between classification and survival explainability strengthens confidence in the identified predictors. Staging variables remained dominant, while socioeconomic indicators also contributed strongly, reinforcing that structural inequities are relevant to prognosis in this setting. This cross-framework consistency also carries an important methodological implication: the outcome is predictably structured both as event occurrence (binary mortality) and as event timing (time-to-event). In other words, the same underlying risk signal remains stable across distinct modeling tasks, supporting the clinical robustness of the prognostic information extracted from registry-linked administrative data.

These findings support the use of administrative-data ML models as pragmatic risk-stratification tools within SUS, particularly where biomarker completeness is limited. External validation across additional Brazilian states and prospective implementation testing remain priorities.

5 Conclusions

Machine learning models trained on linked RHC–SIM data can provide clinically meaningful prognostic stratification for prostate cancer in Brazil. LightGBM was the best classification model, and Gradient Boosting Survival Analysis was the best survival model, using 16 administratively derived predictors.

The combination of predictive performance and explainability supports translational use in public health workflows, including identification of high-risk patients and prioritization of follow-up. Future work should focus on external validation, calibration monitoring, and prospective clinical impact evaluation.

Data Availability Statement

The data underlying this study contain potentially identifying patient information and cannot be made freely available due to ethical and legal restrictions under the Brazilian General Data Protection Law (Lei Geral de Protecao de Dados, LGPD). Researchers may request access to de-identified datasets through [PLACEHOLDER: institutional data access committee and contact information].

Code Availability

The complete analysis pipeline, including preprocessing, feature selection, model training, evaluation, and explainability code, is available at <https://github.com/fabianofilho/ai-prostate-cancer-2026>.

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Competing Interests

The authors declare no competing interests.

Author Contributions

FNBF: Conceptualization, methodology, software, formal analysis, data curation, writing—original draft, visualization. WRG: Conceptualization, methodology, writing—review and editing. VFV: Conceptualization, writing—review and editing. LCLJ: Conceptualization, supervision, writing—review and editing.

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Appendix: Supplementary Figures

The following supplementary figures provide additional model diagnostics and secondary visualizations.

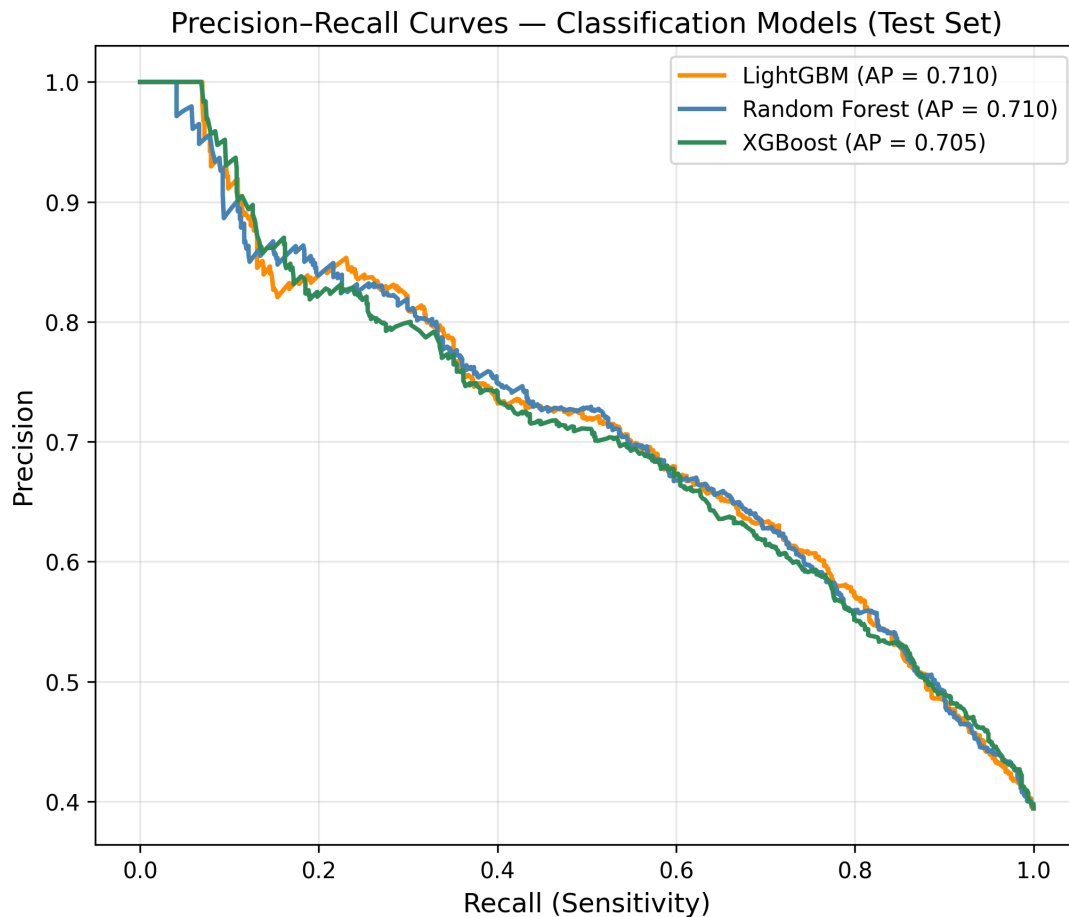


Figure 5: Precision–recall curves across 3 models (LightGBM, Random Forest, and XGBoost) on the test set. Average precision (AP) shown in legend.

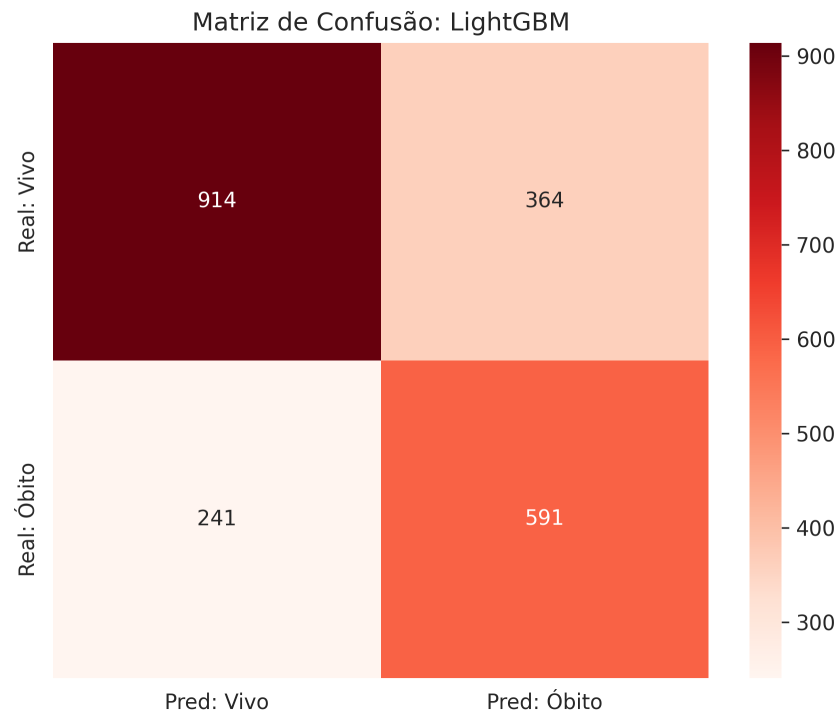


Figure 6: Confusion Matrix: LightGBM.

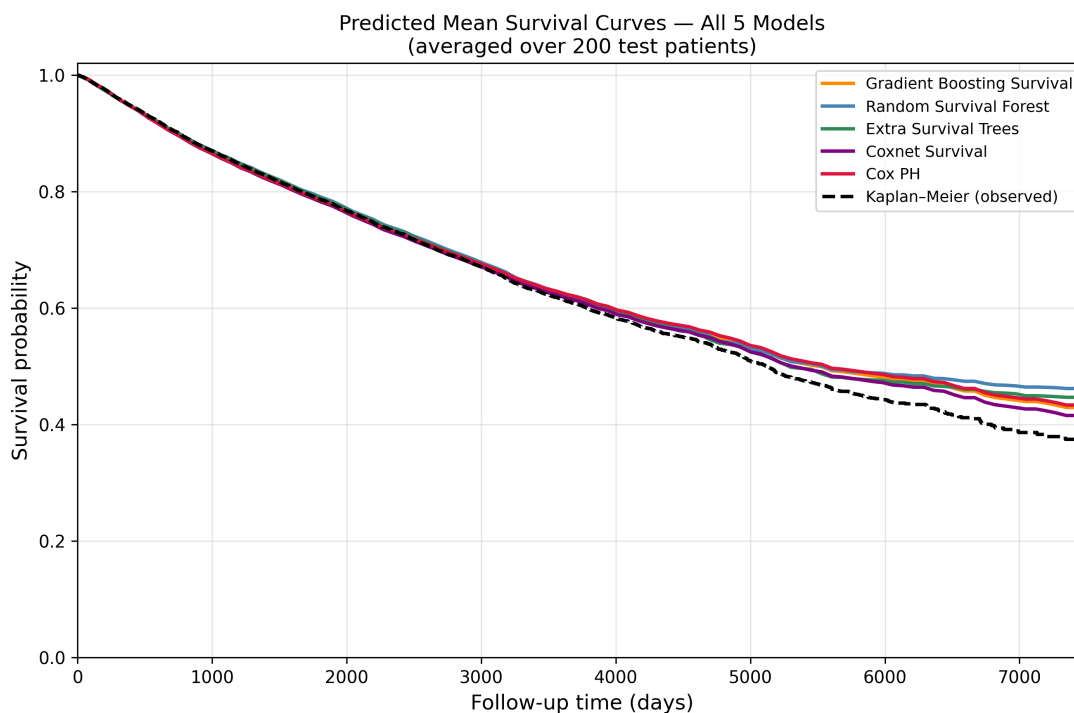


Figure 7: Comparison of survival curves across five models: Gradient Boosting Survival (C-index=0.7093), Random Survival Forest (0.7070), Coxnet (0.7059), Cox PH (0.7046), and Extra Survival Trees (0.7045).

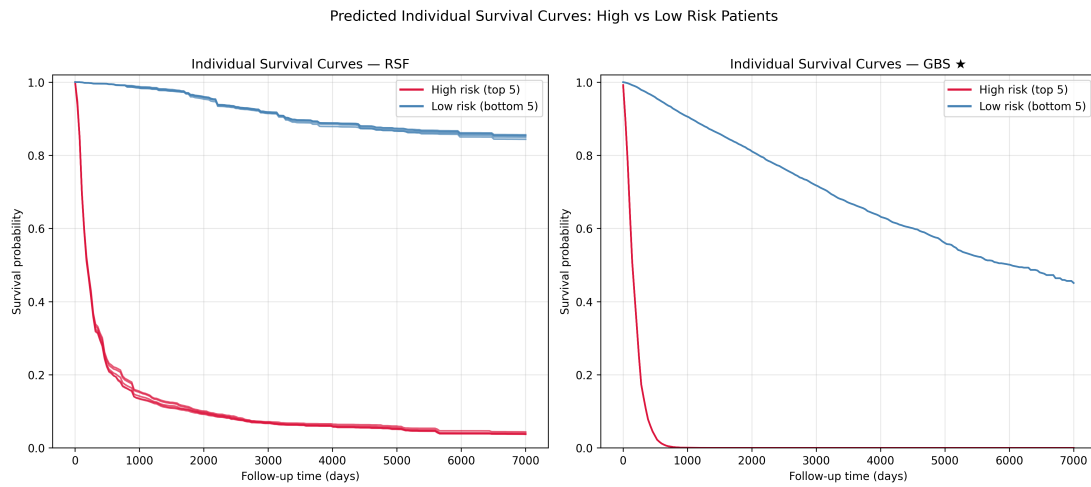


Figure 8: Individual survival trajectories stratified by predicted risk groups.

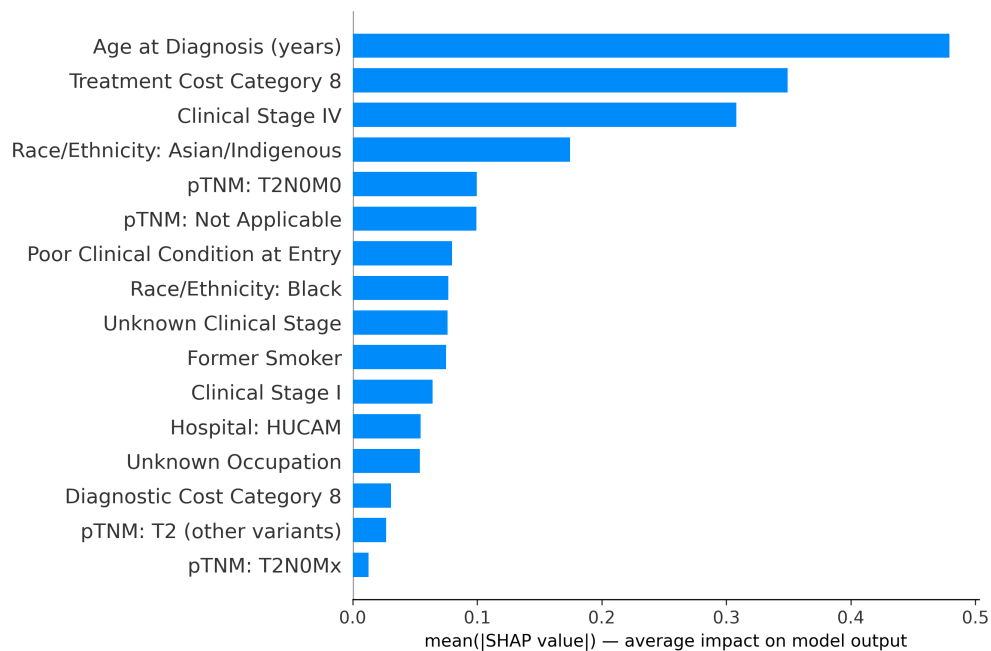


Figure 9: Global SHAP feature importance ranking.